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### PRACTICE5: BASIC LINUX CMD

```
aditya@DESKTOP-FNI4D15:~$ pwd
/home/aditya
aditya@DESKTOP-FNI4D15:~$ ls
UERANSIM  agw_install_docker.sh  aws  awscli2.zip  magma  snap
aditya@DESKTOP-FNI4D15:~$ mkdir UNIT2_LAB
aditya@DESKTOP-FNI4D15:~$ cd UNIT2_LAB
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ pwd
/home/aditya/UNIT2_LAB
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ touch data.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ ls
data.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cat data.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ echo "This is some sample data" >> data.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cat data.txt
This is some sample data
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ ls > filelist.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cat filelist.txt
data.txt
filelist.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cat
Hello
Hello
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ echo "Hello Ubuntu"
Hello Ubuntu
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cat abc.txt
cat: abc.txt: No such file or directory
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ grep "data" data.txt
This is some sample data
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ grep -v "data" data.txt
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ grep "data" data.txt | grep "sample"
This is some sample data
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cut -d " " -f2 data.txt
is
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ cut -c1-5 data.txt
This
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ awk '{print $2}' data.txt
is
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ awk 'END {print NR}' data.txt
1
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$ grep "data" data.txt | cut -d " " -f2 | awk '{print substr($1,1,3)}'
is
aditya@DESKTOP-FNI4D15:~/UNIT2_LAB$
```

### VLE2:

```
aditya@DESKTOP-FNI4D15:~$ aws --version
aws-cli/2.36.11 Python/3.14.6 Linux/6.6.87.1-microsoft-standard-WSL2 exe/x86_64.ubuntu.24
aditya@DESKTOP-FNI4D15:~$ pwd
/home/aditya
aditya@DESKTOP-FNI4D15:~$ ls
IT-Lab-03  UERANSIM  UNIT2_LAB  agw_install_docker.sh  aws  awscli2.zip  magma  snap
aditya@DESKTOP-FNI4D15:~$ aws configure

Tip: You can deliver temporary credentials to the AWS CLI using your AWS Console session by ru

AWS Access Key ID [None]: AKIAR05KX06YUW54140Y
AWS Secret Access Key [None]: 9R8j3Pm5yX+as4uH-578G-eZNEw2mAkQnGChFQS
AWS Session Token [None]: IQoJb3JpZ2ZlLuX2VjEML////////wEaCXVzLXdlc3QtMiJHMEUCIQDoJulhWcUNJKOC
tphnOjilYhWG7AkAWJZJd3ClVMipfkiwZ1i+fGBhRfcY3AHpWt+hkPjCUXUEof1C
Default region name [None]: us-east-1
Default output format [None]: json
aditya@DESKTOP-FNI4D15:~$ |
```

```

aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ aws sts get-caller-identity
{
  "UserId": "AROAR5YXBQXKXVGA3MGJR:user5276813=Aditya_Kumar_Singh",
  "Account": "132655449557",
  "Arn": "arn:aws:sts::132655449557:assumed-role/voclabs/user5276813=Aditya_Kumar_Singh"
}
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ aws --version
aws-cli/2.36.11 Python/3.14.6 Linux/6.6.87.1-microsoft-standard-WSL2 exe/x86_64.ubuntu.24
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ aws configure list
NAME      : VALUE                                : TYPE                : LOCATION
profile   : <not set>                             : None                : None
access_key : *****WXDY                          : shared-credentials-file :
secret_key : *****FQS                           : shared-credentials-file :
region    : us-east-1                             : config-file         : ~/.aws/config

```

## Scenario 1 — Student Records Folder Setup

Your department maintains student academic records in the Linux system. You are placed in the IT-Lab-03 folder and asked to organize files.

Tasks:

- Check what files exist in your current folder.
- Move into the folder named Sem-6.
- Create a new empty file named records.txt.
- View the content of records.txt (verify it is empty).

```

aditya@DESKTOP-FNI4D15:~$ mkdir -p IT-Lab-03/Sem-6
aditya@DESKTOP-FNI4D15:~$ ls -la IT-Lab-03/
total 12
drwxr-xr-x  3 aditya aditya 4096 Aug  9 17:46 .
drwxr-x--- 13 aditya aditya 4096 Aug  9 17:46 ..
drwxr-xr-x  2 aditya aditya 4096 Aug  9 17:46 Sem-6
aditya@DESKTOP-FNI4D15:~$ cd IT-Lab-03/Sem-6
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ pwd
/home/aditya/IT-Lab-03/Sem-6
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ touch records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ ls -la
total 8
drwxr-xr-x 2 aditya aditya 4096 Aug  9 17:46 .
drwxr-xr-x 3 aditya aditya 4096 Aug  9 17:46 ..
-rw-r--r-- 1 aditya aditya   0 Aug  9 17:46 records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ |

```

## Scenario 2 — Appending Results from Terminal

A faculty member gives the following student details to be appended into records.txt:

RegNo: 1023 | Name: Arjun | Grade: A

Tasks:

- Append the above line to records.txt using a redirection operator.
- Now overwrite the entire records.txt so that it contains only: RegNo: 1024 | Name: Ravi | Grade: B

- Create a file named filelist.txt that stores the output of listing all the files in the current directory.

```
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ echo "RegNo: 1023 | Name: Arjun | Grade: A" >> records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat records.txt
RegNo: 1023 | Name: Arjun | Grade: A
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ echo "RegNo: 1024 | Name: Ravi | Grade: B" > records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat records.txt
RegNo: 1024 | Name: Ravi | Grade: B
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ ls -la > filelist.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat filelist.txt
total 12
drwxr-xr-x 2 aditya aditya 4096 Aug  9 17:49 .
drwxr-xr-x 3 aditya aditya 4096 Aug  9 17:46 ..
-rw-r--r-- 1 aditya aditya   0 Aug  9 17:49 filelist.txt
-rw-r--r-- 1 aditya aditya  36 Aug  9 17:49 records.txt
```

## Scenario 3 — Debugging Execution Logs (STDIN, STDOUT, STDERR)

During compilation of a C program, the system produces three different types of messages.

Tasks:

- Describe STDIN, STDOUT, STDERR in your own words.
- Give one example command for each:
  - For reading input from user (STDIN)
  - For printing normal output (STDOUT)
  - For displaying an error (STDERR)

(Example expected: Students may use `cat`, `echo`, invalid commands, etc.)

| Stream        | Meaning         | Simple explanation                                       |
|---------------|-----------------|----------------------------------------------------------|
| <b>STDIN</b>  | Standard Input  | Program ko input dene ka normal source, usually keyboard |
| <b>STDOUT</b> | Standard Output | Program ka normal output, usually terminal               |
| <b>STDERR</b> | Standard Error  | Error/warning messages ko display karne ke liye          |

```
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat > input.txt
Hello Aws!
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ echo "This is normal output (STDOUT)"
This is normal output (STDOUT)
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ invalid_command_xyz
invalid_command_xyz: command not found
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ |
```

## Scenario 4 — Searching Inside Text with grep

The Dean asks you to check whether student records contain the keyword "Grade".

Tasks:

- Search in records.txt for lines containing "Grade".
- Display only those lines that do not contain "A".
- Apply a pipe to:
  - Search lines containing "Grade"
  - Then filter again for "B"

Expected form: `grep "Grade" records.txt | grep "B"`

```
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ echo "RegNo: 1023 | Name: Arjun | Grade: A" >> records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ echo "RegNo: 1025 | Name: Priya | Grade: A" >> records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ echo "RegNo: 1026 | Name: Kumar | Grade: C" >> records.txt
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat records.txt
RegNo: 1024 | Name: Ravi | Grade: B
RegNo: 1023 | Name: Arjun | Grade: A
RegNo: 1025 | Name: Priya | Grade: A
RegNo: 1026 | Name: Kumar | Grade: C
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep "Grade" records.txt
RegNo: 1024 | Name: Ravi | Grade: B
RegNo: 1023 | Name: Arjun | Grade: A
RegNo: 1025 | Name: Priya | Grade: A
RegNo: 1026 | Name: Kumar | Grade: C
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep -v "A" records.txt
RegNo: 1024 | Name: Ravi | Grade: B
RegNo: 1026 | Name: Kumar | Grade: C
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep "Grade" records.txt | grep "B"
RegNo: 1024 | Name: Ravi | Grade: B
```

## Scenario 5 — Extracting Specific Fields using cut

Your file contains data in the format: `RegNo: 1024 | Name: Ravi | Grade: B`

Tasks:

- Extract only the student name (2nd word after spaces) using `cut`.
- Extract the first 5 characters from every line in the file.

```
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cut -d" " -f2 records.txt
1024
1023
1025
1026
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cut d"|" -f2 records.txt
cut: 'd|': No such file or directory
RegNo: 1024 | Name: Ravi | Grade: B
RegNo: 1023 | Name: Arjun | Grade: A
RegNo: 1025 | Name: Priya | Grade: A
RegNo: 1026 | Name: Kumar | Grade: C
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cut -c 1-5 records.txt
RegNo
RegNo
RegNo
RegNo
```

## Scenario 6 — Working with Fields using awk

Your system stores multiple student details inside records.txt.

Tasks:

- Print only the third field (Grade value) from the file.
- Count how many total lines are present in the file using `awk` (Hint: `NR`).

```
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ awk -F'|' '{print $3}' records.txt
Grade: B
Grade: A
Grade: A
Grade: C
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ awk 'END {print NR}' records.txt
4
```

## Scenario 7 — Data Pipeline Automation (Command Chaining)

Suppose `records.txt` contains multiple entries like:

```
RegNo: 1024 | Name: Ravi | Grade: B
RegNo: 1023 | Name: Arjun | Grade: A
RegNo: 1028 | Name: Kiran | Grade: A
```

The principal wants to extract only the first three characters of the student names of those who secured Grade A.

Task:

Construct a single Linux pipeline command that will:

- Filter lines with "A"
- Extract the field containing the name
- Print only the first 3 characters of names

(Expected chain uses `grep` | `cut` | `awk` or variations)

```
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ cat records.txt
RegNo: 1024 | Name: Ravi | Grade: B
RegNo: 1023 | Name: Arjun | Grade: A
RegNo: 1025 | Name: Priya | Grade: A
RegNo: 1026 | Name: Kumar | Grade: C
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep "Grade: A" records.txt
RegNo: 1023 | Name: Arjun | Grade: A
RegNo: 1025 | Name: Priya | Grade: A
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep "Grade: A" records.txt | cut -d'|' -f2
Name: Arjun
Name: Priya
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ "Grade: A" records.txt | cut -d'|' -f2 | awk '{print substr($2,1,3)}'
cut: the delimiter must be a single character
Try 'cut --help' for more information.
Grade: A: command not found
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep "Grade: A" records.txt | cut -d'|' -f2 | awk '{print substr($2,1,3)}'
cut: the delimiter must be a single character
Try 'cut --help' for more information.
aditya@DESKTOP-FNI4D15:~/IT-Lab-03/Sem-6$ grep "Grade: A" records.txt | cut -d '|' -f 2 | awk '{print substr($2,1,3)}'
Arj
Pri
```